



WHAT STUDENTS DO

Each mission gives a spacecraft, a goal (circularize, raise an orbit, escape, transfer, moon flyby) and a **par Δv budget**. Real Newtonian gravity is integrated live, and **the planet is Earth at true scale** ($\mu = 398,600 \text{ km}^3/\text{s}^2$, $R = 6,371 \text{ km}$): speeds in km/s, Δv in m/s, altitudes in km — every number transfers to real missions. Students time-warp, aim a burn, preview the dotted trajectory, commit. Lowest total Δv wins. Runs offline in any browser — open **index.html** (Chromebook-friendly). The moon is fictional (closer/heavier than Luna) so an assist fits one class.

20-MINUTE SESSION PLAN

- 0–3 min** Project the game (⚙️ → Projector mode). Fly Level 1 together: show time-warp, the burn planner, and the dotted preview. Vocabulary: periapsis (low, fast), apoapsis (high, slow), prograde.
- 3–8 min** Students play Levels 1–2 in pairs. Ask them to open the **Physics HUD** and watch “energy gained per 1 Δv ” as the ship swings around.
- 8–15 min** The main event: **Level 3, “Escape Artist.”** Par is 791 m/s, but the tank holds 2,057 — escaping Earth from apoapsis costs $\approx 1,835 \text{ m/s}$, while the same escape from periapsis costs ≈ 661 . Let them discover this. When someone makes par, have them explain *where* they burned and what the HUD showed.
- 15–19 min** Debrief on the projector using a student’s energy-vs-time plot and the “**show the math**” panel below it, which evaluates $\Delta \epsilon = v \cdot \Delta v + \frac{1}{2} \Delta v^2$ with that student’s actual numbers and checks it against the simulation. The burn is a vertical jump in ϵ — taller when v was larger, for the *same* Δv .
- 19–20 min** Post the leaderboard. Levels 4–7 (Hohmann, powered flyby, finite thrust) make excellent homework or a second session.

THE PHYSICS — OBERTH EFFECT

An impulsive burn changes speed by Δv along the velocity. Kinetic energy per unit mass goes from $\frac{1}{2}v^2$ to $\frac{1}{2}(v+\Delta v)^2$, so

$$\Delta KE = v \cdot \Delta v + \frac{1}{2} \Delta v^2$$

The $v \cdot \Delta v$ term dominates: the same Δv buys energy in proportion to how fast you are already moving. On an orbit, speed peaks at periapsis — so a burn there buys the most orbital energy $\epsilon = \frac{1}{2}v^2 - \mu/r$. Gravitational potential is unchanged at the instant of the burn; only the KE jump differs. In the game, the HUD’s “energy per unit Δv ” readout is exactly v , and students can verify the $\approx 2.8\times$ escape-cost gap on Level 3 themselves.

Where does the extra energy come from? Rocket fuel’s energy is fixed, but the useful energy delivered to the *ship* is force $\times$ distance. At high speed the engine acts over a longer distance per second, so more of the propellant’s energy ends up in the ship and less in the exhaust left behind.

Misconception 1: “The burn is better at periapsis because you’re *closer* to the planet.” Proximity is not the mechanism — **speed** is. $\Delta KE = v \cdot \Delta v + \frac{1}{2} \Delta v^2$ contains v , not r . Being deep in the well matters only because falling there is what made the ship fast. A ship moving equally fast far from the planet would gain just as much energy from the same burn.

Misconception 2: “The Oberth effect is the same as a gravity assist.” An **unpowered gravity assist** (Level 5’s moon) steals energy from the *moon’s orbital motion* — no fuel needed, and the moon slows imperceptibly. An **Oberth burn** spends fuel and merely spends it at the most profitable moment. A “powered flyby” combines both.

Misconception 3: “ Δv is like distance — it costs the same anywhere.” Δv *spent* is the same anywhere; the *energy it buys* is not. That is the whole game.

FINITE THRUST (LEVELS 6–7)

Levels 1–5 use impulsive burns; levels 6–7 give the engine limited acceleration (0.055 and 0.013 m/s^2), so Δv is delivered over hours of arc (steering losses ignored). Two lessons emerge: a single burn is cheapest **centered on periapsis** — start early, or the late half happens at lower speed (lit at Pe: 919 m/s to escape; centered: 775) — and with the weak engine one long burn costs $1,498 \text{ m/s}$ while six $\sim 3 \text{ h}$ **perigee kicks** cost 731 . This is why real upper stages escape via repeated perigee kicks.

DISCUSSION QUESTIONS

- Level 3’s energy plot shows a taller jump in ϵ for the same Δv at periapsis. Which term of $\Delta KE = v \cdot \Delta v + \frac{1}{2} \Delta v^2$ explains this?
- The HUD says “energy per unit $\Delta v =$ current speed.” Differentiate $\frac{1}{2}v^2$ to show why.
- Why does a prograde burn at periapsis raise the *apoapsis*, on the far side of the orbit?
- In Level 5, where does the escape energy come from when you fly close behind the moon without burning? What slows down in exchange?
- Real missions (e.g., Parker Solar Probe, Voyager) plan burns at closest approach. Which levels model this, and why does NASA care so much about Δv budgets?
- Could you escape with a radial-out burn? Try it. Why does pushing “straight up” waste fuel?

CLASSROOM LOGISTICS

- Leaderboards** persist per level on each machine; reset them between class periods via ⚙️ Teacher mode → Reset.
- Pars (m/s): L1 712 · L2 1,107 · L3 791 · L4 1,819 · L5 1,424 · L6 838 · L7 823. Verified minimums: L3 escape $\approx 661 \text{ m/s}$ from periapsis vs $\approx 1,835 \text{ m/s}$ from apoapsis.
- Keys: **Space** pause · **B** burn · **Enter** commit · **Z** undo · **X** cut engine · **P/A** warp to Pe/Ap · **H** HUD · **M** briefing · **R** restart. **Undo** rewinds the last burn (also offered after a crash) — students should iterate, not restart.
- For L6–L7, point students at the HUD **time-to-periapsis**, the planner’s **burn duration**, and the **lead** box beside → Pe: those three let them compute an ignition point instead of guessing.